ADITYA JAIN

Sunnyvale, CA 94086 | +1 (323) 986-9260 | adityajn105@gmail.com | linkedin.com/in/adityajn105 | github.com/adityajn105 | adityajain.me

SUMMARY

Software engineer focused on search ranking and machine learning: 3+ years at Salesforce plus 2+ years in data science. Builds BGE reranking, learning-to-rank and classification models, and Spark/Iceberg data pipelines for Salesforce Agentforce enterprise search, with a focus on model and data quality, rigorous evaluation, latency-aware inference, and taking prototypes to production.

TECHNICAL SKILLS

ML / Deep Learning: PyTorch, TensorFlow, Hugging Face Transformers, cross-encoder rerankers (BGE), learning-to-rank (RankSVM, pointwise/pairwise), knowledge distillation, scikit-learn, CUDA GPU training

Search, NLP & LLMs: hybrid lexical/vector retrieval, BM25, query understanding, entity prediction, LLM evaluation (LLM-as-judge), NDCG/MRR, A/B testing

Languages: Python, Java, Scala, SQL | **Data & Distributed:** Spark/PySpark, Iceberg, Airflow, Salesforce Data Cloud, AWS S3

Production: Kubernetes, Docker, gRPC, TensorFlow Serving, feature flags, mTLS, Vault, Splunk, PagerDuty

PROFESSIONAL EXPERIENCE

Salesforce — Software Engineer, Enterprise Search ML & Data **May 2023 – Present**

- **Shipped ranking features and models**: title-containment features and candidate promotion scoped with a 2.9k-query impact analysis; a hybrid keyword/vector weight sweep (**+7.3% NDCG@10**); pairwise learning-to-rank on click logs (MRR@5 0.787, +10–12%).
- **Added personalization signals to ranking**: enriched user profiles from identity resolution and added record-ownership features for CRM results, in the Java ranking service.
- **Led model-quality evaluation for LLM-powered search**: evaluated the LLM query-understanding rollout over 490 queries and flagged regressions before launch. Co-built a 172k-document, 8k-query benchmark (19.5k relevance judgments) and parallel LLM-judged scoring.
- **Built runtime experimentation into the Java ranking service** (feature gating), so A/B tests and per-tenant model or threshold changes ship in seconds instead of hour-long redeploys.
- **Fine-tuned the BGE cross-encoder reranker** with freshness and popularity signals on 80k+ queries: **NDCG@3 +34%, MRR@10 +24%** on 13.7k held-out queries. Built a query-intent-aware variant adding **+7% NDCG@10** and up to +35% MRR on cross-object search.
- **Set the production inference budget** for GA: a candidate-count vs. p95-latency study fixed the reranker at 160 candidates (**+7% NDCG@10 for +120 ms**). Added reranker timeouts and latency/failure metrics by tenant and region.
- **Trained a TensorFlow query-to-entity classifier**: top-1 accuracy 65% → 80%, top-10 79% → 96%. Later retrained it on a **30× larger dataset** (accuracy@3 91.7% → 94.5%). Contributed batch-wise inference to open-source **ml4ir**.
- **Built Search Analytics end to end**, a paid Data Cloud product: Spark/Scala pipelines from an Iceberg lake (~10M rows/org), 49 core-platform Java changes, and backfill/retention frameworks, taken from beta to GA (**380+ customers**). First team at Salesforce to adopt a new ML-pipelines SDK; now in 8 production regions.
- **Built large-scale data-curation pipelines** (Spark → Iceberg → analytics) for Agentforce KPIs and user feedback in every region. Migrated lake access to an Iceberg REST Catalog over mTLS.
- **Prototyped and collaborated across teams**: built a proof of concept for cross-region S3 writes to replace bucket replication, working with 6+ partner teams and architects. Presented model cards, evaluations and A/B-testing proposals to senior leadership; wrote the team runbook and onboarded engineers onto the codebase.

Taxbit — Software Engineer (Transaction Storage) **Jan 2023 – Apr 2023**

- Built Node.js/PostgreSQL transaction-storage APIs for real-time tax engines on AWS; Locust load testing.

Salesforce — Software Engineer Intern, Einstein Search **May 2022 – Aug 2022**

- Built an ML anomaly-detection framework for search metrics across **1,000+ production orgs**, training time-series models in parallel on Spark over 100k org × metric series (0.98 F1).

USC Institute for Creative Technologies — Research Assistant **Feb 2021 – May 2022**

- Raised grading accuracy to **85%** in a dialog-based tutoring platform with a clustering-based NLP pipeline that solved cold start with only 200–300 labeled samples.

Cognizant Technology Solutions — Associate Projects (Data Science) **Sep 2018 – Jan 2021**

- **+10% click-through rate** for an ad-search engine: distilled a fine-tuned **BERT** teacher into a CLSM model (NDCG +5%) and served it with TensorFlow Serving/gRPC at **5 ms** latency.
- **Medicare Star Analytics**: built analytical and ML models achieving a **0.75 AUC** score to predict patient health decline, enabling targeted outreach programs to improve Medicare plan STAR ratings.

EDUCATION

University of Southern California — M.S. Computer Science (Honors), GPA 4.0/4.0; TA, Applied NLP **Dec 2022**

Maharashtra Institute of Technology, Pune — B.E. Computer Engineering, GPA 3.6/4.0 **May 2018**

AWARDS

FY26 Salesforce merit award · Team hackathon "Get It Shipped" category winner · First Runner-Up, Smart India Hackathon 2017 (250 teams)